

Jayavaram Ranga Mokshagna

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LinkedIn | GitHub | Portfolio

Summary

B.Tech Computer Science Engineering student specializing in Big Data Analytics with a strong foundation in programming, databases, data analytics, and computer science fundamentals. Hands-on experience developing real-world projects using Python, SQL, Machine Learning, and modern software tools. Strong analytical and problem-solving skills with the ability to learn new technologies and adapt to diverse technical roles.

Education

SRM Institute of Science and Technology, B.Tech · Computer Science and Engineering - Big Data Analytics

July 2023 – May 2027

- CGPA: 7.93/10

Technical Skills

Programming Languages: Python, C, SQL

Data & Machine Learning: Pandas, NumPy, Scikit-learn, Machine Learning, Data Analytics

Databases: MySQL, MongoDB

Core Fundamentals: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Computer Networks

Visualization & Development Tools: Power BI, Streamlit, Git, GitHub, Jupyter Notebook, VS Code, Docker

Projects

AquaSense – Real-Time Water Quality Monitoring & Prediction

- Built an ESP32-based IoT system measuring pH, turbidity, and temperature every 5 seconds via WiFi.
- Compared 6 ML models on 31,103 real water-quality samples and selected a pruned Decision Tree with 99.79% accuracy (F1=0.9979).
- Developed a React dashboard with WebSocket updates, threshold alerts, manual prediction, and CSV batch upload.
- Containerized 4 services using Docker Compose for one-command deployment.

ClimateQ – Climate-Driven Displacement Risk Predictor

- Merged 5 international datasets (NASA, UNHCR, EM-DAT, World Bank, Copernicus) into 8,961 records across 194 countries from 3 GB NetCDF climate files using Python and xarray.
- Compared 7 ML models and selected XGBoost to predict climate migration risk into 4 levels (Low, Medium, High, Extreme).
- Built animated Plotly choropleth maps, Folium hotspot bubble maps, and an interactive Power BI dashboard showing 120 years of global displacement trends.

Inbox Insights – Exploratory Data Analysis on Personal Email

- Analyzed 26,871 Gmail messages spanning 4.6 years using Python and Pandas.
- Fixed a date-parsing bug that recovered 99% of dropped records and identified key email usage trends.
- Found that 92.7% of received emails were unopened through statistical analysis.

Certifications

Introduction to Machine Learning – NPTEL
Credential ID: NPTEL25CS149S638501118

Sep 2025

Data Analysis with Python – Coursera

Apr 2026